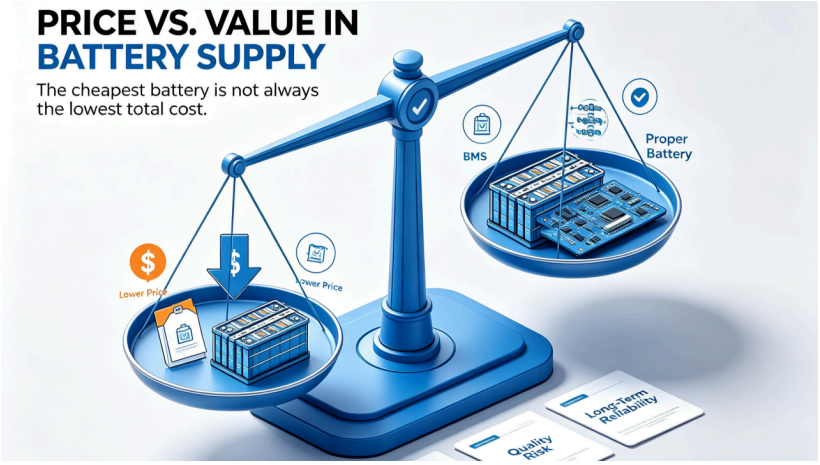
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The Battery Price War Will Not Last Forever

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Battery Solutions for AMR, AGV, UGV & Mobile Robotics | Custom Li-ion Battery Packs | OEM/ODM | BMS & CAN...

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For the past few years, one question has appeared in almost every battery project: "Can you give us a better price?"

I understand why.

Battery prices have fallen dramatically. In 2025, the global average lithium-ion battery pack price reached a new low of around $108/kWh, according to BloombergNEF. Prices in China were even lower, at around $84/kWh.

For buyers, this is good news.

But after working with customers on forklift, AGV, UGV, UAV and industrial battery projects, I have started to think about another question:

How much longer can the industry compete mainly on price?

I don't think the answer is "forever."

The first stage was about making batteries cheaper

The battery industry has done something remarkable over the past decade.

Manufacturers have increased production capacity, improved yields, optimized materials, scaled up supply chains and moved toward lower-cost chemistries such as LFP.

Competition has also become extremely intense.

That combination has pushed prices down very quickly.

And honestly, this competition has been necessary.

Without it, many applications that are becoming common today — electric vehicles, energy storage, forklifts, AGVs and other industrial equipment — would have been much more expensive.

But there is a point where lower prices stop coming mainly from innovation and start coming from margin pressure.

That is where I think the industry is getting increasingly interesting.



A cheaper battery is not necessarily a cheaper solution

This is something I see quite often in custom battery projects.

Two suppliers may quote:

48V 230Ah

Both may say:

- LiFePO4
- Grade A cells
- Smart BMS
- CAN communication
- IP65/IP67
- 5-year life

On paper, they look almost identical.

But the real battery may be very different.

Cell consistency.

BMS configuration.

Busbar design.

Thermal management.

Connector quality.

Communication protocol.

Mechanical structure.

Testing.

Low-temperature protection.

Peak current capability.

And, perhaps most importantly, how the battery behaves after months of real operation.

A battery is not just a collection of cells.

It is a system.

And when that system is installed inside a forklift or AGV that works every day, **downtime can cost much more than the battery itself.**

This is why I believe the industry will gradually move from:

"How much does the battery cost?"

to:

"How much does the battery cost over its entire working life?"

The next competition will be about Total Cost of Ownership

Imagine two batteries.

Battery A is 10% cheaper.

Battery B is 10% more expensive.

If Battery A provides slightly lower usable energy, requires more maintenance, has more protection events, or needs replacement earlier, the initial saving may disappear very quickly.

For an industrial customer, the calculation is not simply:

Battery price = $X

It should be closer to:

Purchase price + charging cost + maintenance + downtime + replacement + service

That is the real cost.

This is especially important for equipment such as forklifts and AGVs.

A battery that stops working in a private garage is one problem.

A battery that stops an automated warehouse line is a completely different problem.



And I think something else is coming

As battery prices continue to fall, **the battery itself will become less differentiated by basic specifications.**

48V will still be 48V.

230Ah will still be 230Ah.

LiFePO4 will still be LiFePO4.

The real differentiation will move deeper into the system.

Smart BMS.

Remote monitoring.

SOC/SOH accuracy.

Predictive maintenance.

CAN communication.

OTA updates.

Thermal management.

Data traceability.

Battery lifecycle management.

Even the ability to understand the machine that the battery is powering.

This is why I don't think the future battery supplier will simply be a company that can assemble cells into a box.

The better suppliers will increasingly become **engineering partners**.

Price will still matter

I don't want to give the impression that price will suddenly become unimportant.

It won't.

Customers will always compare prices.

And competition will continue to push manufacturers to improve efficiency.

In fact, battery prices may continue to decline in the near term. BloombergNEF expects further price reductions in 2026, while IEA has also highlighted continued cost pressure from competition and manufacturing scale.

But I think the market will become more divided.

At one end:

Standardized batteries → highly price-driven

At the other:

Customized industrial batteries → performance, reliability, integration and lifecycle cost

And the second category is where I see a lot of opportunity.

The interesting part of the next 5 years

I don't think the winner will simply be the company offering the lowest price.

I think the winners will be the companies that can answer questions like:

Can this battery communicate with our AGV?

Can it handle our real peak current?

What happens at -20°C?

How will the BMS behave after five years?

Can we monitor the battery remotely?

Can you modify the battery mechanically for our machine?

Can you provide the data we need for certification and compliance?

What happens if something goes wrong in the field?

Those questions are much harder to answer than:

“How much is it per kWh?”

And that is exactly why they matter.

I don't think the battery price war is ending.

I think it is changing.

The first battle was:

Who can make batteries cheaper?

The next battle may be:

Who can make batteries cheaper without making the entire system more expensive?

And eventually:

Who can create the most value from every kilowatt-hour?

For me, that is a much more interesting future.

Because the cheapest battery will always have a market.

But the battery that helps a forklift run longer, an AGV avoid downtime, a UGV perform reliably in harsh environments, or an industrial fleet predict failures before they happen —

that battery is no longer just a commodity.

It becomes part of the machine.

And once the battery becomes part of the machine, **price is only one part of the conversation.**

#BatteryIndustry #LithiumBatteries #BatteryTechnology #EnergyStorage #IndustrialBatteries #AGV #AMR #Forklift #Robotics #Forklift #UGV

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